

RTS16 – THE RTS EQUATION: THE ONE-INCH EQUATION OF THE UNIVERSE

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Abstract

This episode presents, for the first time in its complete form, the fundamental equation of the Primary Network, as derived from RTS and VG. It is a short, elegant equation, whose simplicity contrasts with the depth of its implications. It governs the dynamics of the entire Network – from the individual node to the cosmos – and unifies, in a single expression, gravity, matter, time, and the nature of information. The equation builds on the idea that spacetime curvature is the macroscopic expression of **resonance between nodes**, while gravity is an emergent effect of this resonance, not a fundamental force.

Keywords: fundamental equation, coupled oscillators, Primary Network, unification, VG, RTS.

1. INTRODUCTION – AN EQUATION SOUGHT FOR A LIFETIME

Michio Kaku, one of the most influential figures in modern theoretical physics and a co-founder of String Theory, has often expressed a deep desire to find a "**one-inch equation**" – a short, elegant mathematical expression that would govern the entire universe. The RTS equation we propose is **that equation**. It is simple, elegant and builds on the deepest insights of modern physics while opening a new path.

2. THE RTS EQUATION – BASIC FORM

The fundamental equation of the Primary Network is:

$$H = \sum_i \left(\frac{p_i^2}{2} + \frac{1}{2} \omega_i^2 q_i^2 \right) + \sum_{i < j} K_{ij} q_i q_j$$

Meaning of each term:

- **H** – The **total Hamiltonian** of the Primary Network. All laws of physics emerge from it.
- **p_i and q_i** – The **canonical variables** of node i . q_i represents the **deformation of the cavity** (deviation from spherical shape), and p_i is the **associated momentum**. Together, they describe the **vibrational state** of the node.
- **ω_i** – The **natural frequency** of node i , determined by the spins of its links.
- **K_{ij}** – The **coupling constant** between nodes i and j , determined by the spin of the link connecting them. This **term allows direct coupling between any two nodes in the Network, regardless of distance**.

This equation states:

Everything is vibration – each node vibrates on its own, but all vibrations are connected to one another, regardless of distance.

3. WHAT THIS EQUATION GOVERNS – FIVE ESSENTIAL POINTS

3.1. Gravity – In the continuum limit (as the number of nodes $N \rightarrow \infty$ and the distance between them $\rightarrow 0$), this equation reduces to **Einstein's equations**. Gravity is not a fundamental force, but an **emergent effect** of the resonance between nodes. Spacetime curvature is the macroscopic expression of this resonance.

3.2. Matter – Local excitations (high-frequency normal modes) correspond to **elementary particles**. Each particle is a **vibration** of the Network.

3.3. Time – Time is an **emergent property** of coupling dynamics. It flows forward during expansion and backward during contraction.

3.4. Information – Information is encoded in the vibration patterns of nodes, carried by nodal waves and stored in the subquantum wall. The Universe has no DELETE function – all information is preserved in the spiral-matrix.

3.5. The Torsion Field (CT) – As shown in VG9, the torsion field is the fundamental force, and gravity is its emergent manifestation at the cosmic scale. The RTS equation is fully compatible with this view.

4. CONCLUSION – THE "ONE-INCH EQUATION"

This is the **"one-inch equation"** we propose. It is short, elegant and governs everything – from particles to gravity, from time to information.

With this equation, RTS becomes a **unified theory of reality**, embracing both quantum physics and cosmology.

CORRESPONDENCE BETWEEN RTS16 AND PREVIOUS EPISODES

<i>RTS16 Point</i>	<i>Topic</i>	<i>Main Sources</i>
<i>3.1. Gravity</i>	Gravity as an emergent effect of oscillator couplings.	VG7 (Nodal Time and Gravity – A Fundamental Relation); RTS4 (Time as a Nodal Relation)
<i>3.2. Matter</i>	Elementary particles as local excitations (high-frequency normal modes) of the Network.	RTS6 (Nodal Engineering); RTS9 (Fundamentals of Nodal Engineering); VG8 (The Architecture of Information in the Primary Network)
<i>3.3. Time</i>	Time as an emergent property of coupling dynamics; reversal during contraction.	RTS4 (Time as a Nodal Relation); VG5 (The Dance of the Network – From Quantum Foam to Spiral Disc); discussion on spin and contraction
<i>3.4. Holography</i>	All volume excitations are determined by surface excitations; 3D space as a projection.	RTS15 (Vibrational Foundations of the Network – From Node to Hologram); VG8 (The Architecture of Information in the Primary Network)
<i>3.5. Information</i>	Information encoded in vibration patterns; nodal waves; subquantum wall; "The Universe has no DELETE function."	VG8 (The Architecture of Information in the Primary Network); RTS14 (Taxonomy of Inter-Slice Interactions); discussion on Akasha and the subquantum wall